

Gaussian drift separates the intrinsic semigroup BMO norms

Automated mathematics run `fa_banach_001`

Lane 10 — candidate counterexample for expert review

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Status. Candidate counterexample, likely valid. For the classical one-dimensional Ornstein–Uhlenbeck semigroup, bounded coordinate truncations have uniformly bounded $\text{bmo}(\mathcal{T})$ norm while their $\text{BMO}(\mathcal{T})$ norms and $\sup_t \|T_t f - T_{2t} f\|_\infty$ diverge. The coordinate also separates $\text{BMO}(\Gamma)$ from $\text{BMO}(\mathcal{P})$ and $\text{BMO}(\partial)$. This answers the universal readings of open problem (ii)(a),(c),(c') of Junge–Mei negatively, but does not characterize all semigroups for which the comparisons hold.

1 Source question and definitions

Let $\mathcal{T} = (T_t)_{t \geq 0}$ be a standard Markov semigroup. In the commutative case, the two norms in open problem (ii)(c) of Junge and Mei [1] are

$$\|f\|_{\text{bmo}(\mathcal{T})}^2 = \sup_{t > 0} \| |T_t f|^2 - |T_{2t} f|^2 \|_\infty, \quad (1)$$

$$\|f\|_{\text{BMO}(\mathcal{T})}^2 = \sup_{t > 0} \| |T_t f - T_{2t} f|^2 \|_\infty. \quad (2)$$

They ask for conditions under which these norms are equivalent, and single out the estimate

$$\sup_{t > 0} \|T_t f - T_{2t} f\|_\infty \leq C \|f\|_{\text{bmo}(\mathcal{T})}. \quad (3)$$

Part (a) asks for comparison of the Poisson-semigroup norm $\text{BMO}(\mathcal{P})$, the vertical square-function norm $\text{BMO}(\partial)$, and the carré-du-champ norm $\text{BMO}(\Gamma)$.

We prove that even the strongest standard regularity package does not imply (a), (c), or (c'). Work on the probability space

$$d\gamma(x) = (2\pi)^{-1/2} e^{-x^2/2} dx$$

with the Ornstein–Uhlenbeck semigroup

$$T_t f(x) = \mathbb{E} f\left(e^{-t}x + \sqrt{1 - e^{-2t}} Z\right), \quad Z \sim N(0, 1). \quad (4)$$

It is a symmetric, ergodic, analytic and hypercontractive Markov semigroup. For its positive generator $A = -\partial_x^2 + x\partial_x$, $\Gamma(f) = |f'|^2$ and

$$\Gamma_2(f) = |f''|^2 + |f'|^2 \geq \Gamma(f).$$

Thus the example satisfies strictly more than the $\Gamma_2 \geq 0$ hypothesis used in the source.

geometric structure of Euclidean spaces is essential. Mei (see [Mei08](#)) provides a first approach of this problem in the context of von Neumann algebras with two additional assumptions on the semigroups. The authors expect a more general H^1 -BMO duality in the context of semigroups.

(ii) *Comparison of different semigroup BMO-norms.* There are several natural semigroup BMO norms as introduced in this article. A complete comparison of them is in order. In particular, it will be interesting to investigate the conditions on the semigroups so that we have the estimates,

- (a) $\|\cdot\|_{BMO^c(\mathcal{P})} \simeq \|\cdot\|_{BMO^c(\partial)} \simeq \|\cdot\|_{BMO^c(\Gamma)}.$
- (b) $\|\cdot\|_{bmo^c(\mathcal{P})} \simeq \|\cdot\|_{BMO^c(\mathcal{P})} \simeq \|\cdot\|_{BMO^c(\hat{\Gamma})}.$
- (c) $\|\cdot\|_{bmo^c(\mathcal{T})} \simeq \|\cdot\|_{BMO^c(\mathcal{T})}.$
- (c') $\sup_t \|T_t x - T_{2t} x\| \leq c \|x\|_{bmo^c(\mathcal{T})}.$
- (d) $\sup_t \|T_t \int_0^t |\frac{\partial T_s x}{\partial s}|^2 ds\| \leq c \|x\|^2.$

(iii) The classical BMO functions φ on $\mathbb{R}$ is integrable with respect to $\frac{1}{1+t^2}dt$. What is a noncommutative analogue of this property? A more precise question is, does there exist a normal faithful state τ on $\mathcal{N}$ such that $\tau|x| \leq c \|x\|_{BMO(\mathcal{T})}$ for $x \in \mathcal{N}$.

REFERENCES

Figure 1: Open problem (ii) on printed page 33 of the source paper [1].

2 Proof intuition

Mehler's formula turns the expression in (1) into the conditional variance of f under a Gaussian transition kernel. The truncations $f_R(x) = \max(-R, \min(x, R))$ are 1-Lipschitz, so their transition variances never exceed the variance $1 - e^{-2t}$ of the Gaussian input. Their $bmo(\mathcal{T})$ norms are therefore uniformly bounded.

The other norm sees something that conditional variance misses: the OU drift of the mean. On the untruncated coordinate $h(x) = x$, the mean changes from $e^{-t}x$ to $e^{-2t}x$, an unbounded function of x . Since $f_R \rightarrow h$ in $L_2(\gamma)$, the same mean-drift term for f_R must have unbounded L_∞ norm as $R \rightarrow \infty$. Kadison's inequality puts this drift term below the $BMO(\mathcal{T})$ norm. This proves both failures using bounded functions.

For the subordinated Poisson semigroup, the carré du champ sees only the bounded derivative of the coordinate, while the oscillation and vertical derivative norms still see its unbounded drift. That is the mechanism behind the separation in part (a).

3 The bounded counterexample to (c) and (c')

For $R > 0$, set

$$f_R(x) = (-R) \vee x \wedge R. \tag{5}$$

Theorem 1. *For the Ornstein–Uhlenbeck semigroup (4):*

(i) $\|f_R\|_{bmo(\mathcal{T})} \leq 1$ for every $R > 0$;

(ii) for every fixed $t > 0$,

$$\|(T_t - T_{2t})f_R\|_\infty \longrightarrow \infty \quad (R \rightarrow \infty);$$

(iii) consequently $\|f_R\|_{BMO(\mathcal{T})} \rightarrow \infty$.

Hence (3) is false and the two norms in open problem (ii)(c) are not equivalent, even on bounded functions for one fixed classical semigroup satisfying $\Gamma_2 \geq \Gamma$.

Proof. Fix $t > 0$ and $x \in \mathbb{R}$, and let

$$Y = e^{-t}x + \sqrt{1 - e^{-2t}} Z.$$

The quantity in (1) is

$$T_t f_R^2(x) - (T_t f_R(x))^2 = \text{Var}(f_R(Y)).$$

If Y' is an independent copy of Y , then the 1-Lipschitz property gives

$$\text{Var}(f_R(Y)) = \frac{1}{2} \mathbb{E} |f_R(Y) - f_R(Y')|^2 \leq \frac{1}{2} \mathbb{E} |Y - Y'|^2 = 1 - e^{-2t} \leq 1.$$

Taking the two suprema proves (i).

Let $h(x) = x$. Since $f_R \rightarrow h$ in $L_2(\gamma)$ and every T_s is an L_2 contraction,

$$(T_t - T_{2t})f_R \longrightarrow (e^{-t} - e^{-2t})h \quad \text{in } L_2(\gamma). \quad (6)$$

The limit is essentially unbounded. If the L_∞ norms on the left did not tend to infinity, some subsequence would be bounded in L_∞ by one constant. An L_2 -convergent sub-subsequence converges almost everywhere, forcing the limit in (6) to have the same L_∞ bound, a contradiction. This proves (ii).

Finally, positivity and Kadison's inequality give, pointwise,

$$\begin{aligned} |T_t f_R - T_{2t} f_R|^2 &= |T_t(f_R - T_t f_R)|^2 \\ &\leq T_t |f_R - T_t f_R|^2. \end{aligned}$$

Thus $\|(T_t - T_{2t})f_R\|_\infty \leq \|f_R\|_{\text{BMO}(\mathcal{T})}$, and (iii) follows. $\square$

The separation also occurs at the level of the natural L_2 spaces, without truncation.

Proposition 1. For $h(x) = x$,

$$\|h\|_{\text{bmo}(\mathcal{T})} = 1, \quad \|h\|_{\text{BMO}(\mathcal{T})} = \infty.$$

In particular, on centered $L_2(\gamma)$ elements one has a strict inclusion $\text{BMO}(\mathcal{T}) \subsetneq \text{bmo}(\mathcal{T})$.

Proof. Mehler's formula gives

$$T_t h = e^{-t} h, \quad T_t h^2 = e^{-2t} h^2 + 1 - e^{-2t}.$$

The first assertion follows immediately from (1). On the other hand,

$$T_t |h - T_t h|^2 = (1 - e^{-t})^2 (e^{-2t} h^2 + 1 - e^{-2t}),$$

which is unbounded for every $t > 0$. The source proves under $\Gamma_2 \geq 0$ that $\text{bmo} \lesssim \text{BMO}$ [1, Proposition 2.4], so the inclusion holds; the coordinate makes it strict. $\square$

4 Separation of the norms in part (a)

Let $P_s = e^{-s\sqrt{A}}$ be the Poisson semigroup subordinated to T . The source defines

$$\begin{aligned}\|f\|_{\text{BMO}(\Gamma)}^2 &= \sup_{t>0} \left\| P_t \int_0^t \Gamma(P_s f) s ds \right\|_\infty, \\ \|f\|_{\text{BMO}(\partial)}^2 &= \sup_{t>0} \left\| P_t \int_0^t |\partial_s P_s f|^2 s ds \right\|_\infty.\end{aligned}$$

Theorem 2. *For $h(x) = x$,*

$$\|h\|_{\text{BMO}(\Gamma)} = \frac{1}{2}, \quad \|h\|_{\text{BMO}(\mathcal{P})} = \|h\|_{\text{BMO}(\partial)} = \infty.$$

Furthermore,

$$\sup_R \|f_R\|_{\text{BMO}(\Gamma)} \leq \frac{1}{2}, \quad \|f_R\|_{\text{BMO}(\mathcal{P})} \longrightarrow \infty.$$

Thus symmetry, ergodicity, analyticity, hypercontractivity and $\Gamma_2 \geq \Gamma$ do not suffice for the triple equivalence in open problem (ii)(a), and the failure is visible through bounded approximants.

Proof. The coordinate lies in the first Wiener chaos: $Ah = h$. Also $A(h^2 - 1) = 2(h^2 - 1)$. Hence

$$P_s h = e^{-s} h, \quad P_t h^2 = 1 + e^{-\sqrt{2}t} (h^2 - 1). \quad (7)$$

Since $\Gamma(P_s h) = e^{-2s}$,

$$\|h\|_{\text{BMO}(\Gamma)}^2 = \sup_{t>0} \int_0^t e^{-2s} s ds = \frac{1}{4}.$$

But

$$P_t |h - P_t h|^2 = (1 - e^{-t})^2 P_t h^2$$

is unbounded by (7); this is the $\text{BMO}(\mathcal{P})$ assertion. Similarly,

$$P_t \int_0^t |\partial_s P_s h|^2 s ds = \left(\int_0^t e^{-2s} s ds \right) P_t h^2$$

is unbounded, proving the $\text{BMO}(\partial)$ assertion.

For the bounded truncations, subordination and the Mehler derivative identity give

$$|(P_s f_R)'| \leq \int_0^\infty \eta_s(u) |(T_u f_R)'| du \leq \int_0^\infty \eta_s(u) e^{-u} du = e^{-s},$$

where η_s is the positive subordination density. Therefore $\Gamma(P_s f_R) \leq e^{-2s}$ and $\|f_R\|_{\text{BMO}(\Gamma)} \leq 1/2$.

Finally $f_R \rightarrow h$ in $L_2(\gamma)$ and $(P_t - P_{2t})h = (e^{-t} - e^{-2t})h$. Repeating the L_2 -limit argument in (6) shows $\|(P_t - P_{2t})f_R\|_\infty \rightarrow \infty$ for every fixed $t > 0$. Kadison's inequality for P_t yields

$$\|(P_t - P_{2t})f_R\|_\infty \leq \|f_R\|_{\text{BMO}(\mathcal{P})},$$

and completes the proof. □

5 Verification, scope, and novelty bounds

The proof uses only Mehler’s formula, Gaussian L_2 convergence, positive subordination, and Kadison’s inequality. No computation is needed. The included script `code/verify_ou_truncations.py` evaluates the first two moments of a clipped Gaussian exactly in terms of the normal cdf and checks on a finite grid that the transition variance is at most one while the fixed-time drift grows linearly with R . This is a diagnostic, not evidence for the proof.

The result fully refutes the universal estimates (c) and (c’). It also shows that the strong hypotheses listed in Theorem 2 are insufficient for (a). The broad request for a complete characterization of all semigroups remains open. Part (b) is not challenged: the source already proves its Poisson comparison under $\Gamma_2 \geq 0$.

A bounded novelty search on 9 August 2026 covered the run’s registry, solution, attempt, and proof-gap indexes; exact and close-variant arXiv searches combining the displayed norm comparison, the $T_t - T_{2t}$ term, and Ornstein–Uhlenbeck; the original paper; Mei’s follow-up [2]; the Poisson H^∞ paper [3]; and the later algebraic Calderón–Zygmund treatment of Gaussian semigroup BMO [4]. The follow-up explicitly says the two norms may differ, proves equality under extra assumptions, and notes that OU fails one of those assumptions, but the searched sources did not contain the explicit coordinate/truncation separation above or the part-(a) separation. Novelty is therefore provisional; this was not an exhaustive MathSciNet or zbMATH priority search.

Human-review recommendation. High priority. Verify the source’s domain conventions, the Lipschitz conditional-variance estimate, and the subordination gradient bound. The bounded sequence f_R makes the failure of (c),(c’) independent of whether the source’s completed BMO spaces are defined from L_∞ or L_2 .

References

- [1] M. Junge and T. Mei, *BMO spaces associated with semigroups of operators*, Math. Ann. 352 (2012), 691–743; arXiv:1111.6194. <https://arxiv.org/abs/1111.6194>.
- [2] T. Mei, *An H^1 –BMO duality theory for semigroups of operators*, arXiv:1204.5082 (2012). <https://arxiv.org/abs/1204.5082>.
- [3] T. Ferguson, T. Mei, and B. Simanek, *H^∞ -calculus for semigroup generators on BMO*, Adv. Math. 347 (2019), 408–441; arXiv:1701.06623. <https://arxiv.org/abs/1701.06623>.
- [4] M. Junge, T. Mei, J. Parcet, and R. Xia, *Algebraic Calderón–Zygmund theory*, Adv. Math. 376 (2021), 107443; arXiv:1907.07375. <https://arxiv.org/abs/1907.07375>.